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Time–Frequency Weighted Losses for Phoneme Reconstruction in DNN-Based Speech Enhancement

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Abstract

Conventional training losses for speech enhancement based on the signal-to-distortion ratio (SDR) treat all time–frequency (TF) regions uniformly, overlooking the fine-grained spectral cues that are relevant to specific phoneme intelligibility. We propose a TF weighting framework that modulates the SDR objective based on local speech presence, speech-to-interference ratio (SIR), and spectral flux. By integrating these factors into a differentiable objective, the framework emphasizes TF bins with high speech–noise competition while also accounting for transient cues such as consonant bursts. Experimental results show that our approach improves objective frequency-weighted enhancement metrics, as well as phoneme recognition accuracy, particularly for consonants. Spectral analysis shows better reconstruction of mid-frequency structures at less adverse SIR.

Index Terms: multichannel speech enhancement, phoneme-level reconstruction, time-frequency weighted training loss

1. Introduction

Deep learning–based multichannel speech enhancement (MCSE) has evolved toward end-to-end neural-spatial frameworks that jointly model spectral and inter-channel dependencies [1, 2, 3]. State-of-the-art models leverage temporal convolutional networks and self-attention to optimize spatial filtering. MCSE algorithms are commonly trained using signal-to-distortion ratio (SDR)-based objectives [1]. While SDR-based losses are effective for improving overall signal reconstruction and interference reduction, they implicitly treat all time or time–frequency (TF) regions uniformly. By assigning equal importance to all TF regions, SDR-based losses overlook the uneven perceptual contribution of different speech components, especially in noisy conditions.

From a phonetic standpoint, intelligibility depends critically on specific acoustic cues that are localized in both time and frequency [4, 5]. Consonantal transitions, plosive bursts, fricative noise bands, and formant trajectories carry disproportionate linguistic information compared to steady-state vowel regions [6, 7]. Moreover, masking effects are the strongest when speech and noise magnitudes are comparable, leading to degraded transmission of these cues [8, 9]. In hearing-assistive scenarios with competing sound sources or background noise, preserving transient and mid-frequency speech structures is particularly important [10, 11, 12]. Consequently, training objectives that emphasize regions of strong speech–noise interaction may better reflect perceptual priorities than globally uniform losses [13, 14, 15]. This intuition is also related to glimpsing theories of speech perception, which emphasize the importance of local spectro-temporal regions that remain informative in the presence of masking noise [16].

Monir et al. [15] previously introduced frequency-weighted SDR formulations to improve phoneme-level behavior. These approaches incorporated perceptual band-importance functions and local signal-to-interference (SIR) modulation, yet did not explicitly combine speech presence, speech–noise competition, and transient dynamics within a unified weighting framework.

In this work, we investigate a structured TF-weighting framework grounded in speech–noise competition. We first model the speech-noise competition explicitly through SIR and speech-presence gating. We then extend this formulation to transient phonetic cues via spectral flux. Finally, we examine a data-driven alternative where spectral weights are fully learned. This progression allows us to analyze the respective roles of perceptual inductive bias and learned weighting in phoneme-level enhancement.

2. Methodology

2.1. TF-weighted SDR

Conventional SDR-based objectives treat all TF bins uniformly, although their perceptual relevance strongly depends on local speech–noise interaction [8, 13]. In particular, intelligibility is mainly affected in regions where speech and noise have comparable magnitudes, leading to strong competition and masking [8]. In contrast, bins dominated by clean speech or containing negligible energy contribute less to perceptual errors [13]. Therefore, we introduce a TF-weighted SDR framework that explicitly models local speech–noise competition.

Let $S(f,t)$, $N(f,t)$, and $\widehat{S}(f,t)$ denote the Mel-band magnitudes of the clean speech, noise, and estimated speech signals in the TF domain, respectively. In practice we compute their short-time Fourier transform and group frequencies using a Mel scale. Let $S_{\text{proj}}(f,t)$ denote the orthogonal projection of $\widehat{S}(f,t)$ onto $S(f,t)$, and $E_{\text{dist}}(f,t) = \widehat{S}(f,t) - S_{\text{proj}}(f,t)$ the residual distortion component. We then define the TF-weighted SDR loss is as follows:

$$\mathcal{L}_w = -10\log_{10}\frac{\sum_{f,t} w(f,t)\,|S_{\text{proj}}(f,t)|^2}{\sum_{f,t} w(f,t)\,|E_{\text{dist}}(f,t)|^2}, \qquad (1)$$

where $w(f,t) \geq 0$ controls the contribution of each TF bin. When $w(f,t) = 1$, this formulation reduces to an unweighted TF-domain SDR. Note that it is distinct from a time-domain SDR loss [17], which is used here as baseline and denoted $\mathcal{L}_{\text{T}}$. Monir et al. [15] proposed several weighting strategies based on the local SIR defined as $\text{SIR}(f,t) = 10\log_{10}\frac{|S(f,t)|^2}{|N(f,t)|^2}$. Among these, the best performing weights are $w(f,t) = \text{softmax}(-\log(\text{SIR}(f,t)))$, which yield a loss denoted $\mathcal{L}_{\text{logSIR}}$. While promising, these approaches do not fully account for the speech–noise competition.

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Table 1: *Utterance-level performance for the white noise (WN) and speech-shaped noise (SSN) test sets averaged across input* SIR *levels between* -8 *dB and* 8 *dB. Note that input (FW)-SDR and (FW)-SAR are not reported as they are infinite.*

Loss	WN							SSN						
	SIR	SAR	SDR	FW-SIR	FW-SAR	FW-SDR	STOI	SIR	SAR	SDR	FW-SIR	FW-SAR	FW-SDR	STOI
Input	-2.0	–	–	-6.0	–	–	0.50	-2.0	–	–	-4.8	–	–	0.54
$\mathcal{L}_{\mathrm{T}}$	13.3	**2.4**	**1.7**	5.1	**3.5**	4.7	0.63	14.3	**1.2**	**0.7**	7.5	2.2	4.4	**0.63**
$\mathcal{L}_{\mathrm{logSIR}}$	16.2	1.2	0.9	7.6	3.1	4.8	0.61	14.2	-1.1	-1.6	7.5	2.6	3.9	0.57
$\mathcal{L}_{\mathrm{learn}}$	16.1	1.9	1.6	7.5	3.4	5.0	**0.64**	15.0	0.3	-0.1	8.2	2.2	4.6	0.59
$\mathcal{L}_{\mathrm{SIR\cdot SP}}$	15.1	2.0	1.5	6.6	3.3	**5.2**	0.63	13.4	0.4	-0.2	6.8	1.8	3.6	0.57
$\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$	**17.2**	1.8	1.5	**8.4**	2.8	**5.2**	**0.64**	**15.1**	0.8	0.4	**8.3**	**2.3**	**4.9**	0.61

2.2. Proposed weighting schemes

Intuitively, one desires high weight values when both $|S(f,t)|$ and $|N(f,t)|$ are large, as these bins correspond to strong speech–noise competition. Conversely, bins with dominant speech (high local SIR) or very low speech energy should receive smaller weights in order to avoid over-optimizing perceptually less critical regions. We propose the following schemes that comply with these requirements. The first two schemes are grounded in explicit modeling of speech–noise competition, while the third examines whether a purely learned spectral profile can achieve similar benefits.

SIR–speech presence weighting First, let us define the two following gating functions:

$$g_{\mathrm{SIR}}(f,t) = \sigma(-\mathrm{SIR}(f,t) + \tau_1), \tag{2}$$

$$g_{\mathrm{SP}}(f,t) = \sigma(|S(f,t)|^{\gamma} + \tau_2), \tag{3}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and τ_1 and τ_2 are learned thresholds. We propose the following weights:

$$w(f,t) = g_{\mathrm{SIR}}(f,t) \cdot g_{\mathrm{SP}}(f,t) \tag{4}$$

which emphasizes TF bins where speech remains active despite low SIR. The corresponding loss is denoted $\mathcal{L}_{\mathrm{SIR\cdot SP}}$.

SIR–speech presence–spectral flux weighting When $|S|$ is weak but perceptually relevant, as in several transient phonemes (e.g., plosives), a purely magnitude-based weighting may underestimate their importance. To capture such rapidly varying cues, we consider the frame-wise spectral flux

$$\mathrm{SF}(t) = \frac{\sum_f \max(|S(f,t)| - |S(f,t-1)|, 0)}{\sum_f |S(f,t-1)| + \epsilon}, \tag{5}$$

which measures the relative increase of spectral energy between consecutive frames. We incorporate this quantity into the weighting scheme as follows:

$$w(f,t) = g_{\mathrm{SIR}}(f,t) \cdot g_{\mathrm{SP}}(f,t) \cdot (1 + k\,\sigma(\mathrm{SF}(t))), \tag{6}$$

where k is a scaling factor adjusting the relative importance of the spectral flux. This scheme increases sensitivity to transient phonetic structures while preserving speech–noise competition modeling. The corresponding loss is denoted $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$.

Learnable weights Lastly, we consider the following scheme:

$$w(f) = \mathrm{softmax}(\theta_f), \tag{7}$$

where the spectral parameters θ_f are learned and time-independent. We initialize θ_f with ANSI 1997 band-importance weights [18]. The corresponding loss is denoted $\mathcal{L}_{\mathrm{learn}}$.

3. Experimental protocol

3.1. Acoustic conditions and dataset

We adopt the data generation and acoustic simulation protocol of Monir et al. [15] for training and validation. Clean speech is drawn from LibriSpeech [19], and noise consists of SSN and ecological noise from the Disconoise dataset [20]. Reverberant mixtures are generated using room impulse responses (RIRs) simulated with Pyroomacoustics [21], with randomized room geometry and RT_{60}, and a 4-channel binaural hearing-aid microphone configuration. Mixtures are created over SIR values in $[-10, 10]$ dB. For evaluation, we consider two dedicated test sets: one with WN and one with SSN, computed on 10 speakers (5 male, 5 female). These test sets follow a controlled configuration, with the target speech placed at 0° (front) and the masker at 45° (right), using measured RIRs from the Binaurec dataset [22] at SIR levels from -8 dB to 8 dB.

3.2. Speech enhancement algorithm

As MCSE algorithm we use FaSNet, an end-to-end time-domain two-stage adaptive beamformer [1]. We follow the training process [1] of Monir et al. [15], relying on Asteroid [23]. For a binaural hearing-aid configuration, microphone permutation is disabled to maintain a fixed front-left reference channel. The factor k in $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ is tuned on the validation set to 0.2.

3.3. Performance measures

As objective metrics, we report scale-invariant SDR, SIR, and signal-to-artifact ratios (SAR) computed in the time-domain [24, 17] on the left ear, which is the best one [25]. In addition, we report their intelligibility-oriented frequency-weighted (FW) counterparts [13, 14], with weights proportional to $|S(f,t)|^{0.2}$ [14, 15]. We also include short-time objective intelligibility (STOI) [26] to assess intelligibility.

We additionally report the word error rate (WER) and phoneme accuracy (PA) when performing speech and phoneme recognition after MCSE using Wav2Vec2-based models [27, 28]. PA is defined as the percentage of correctly predicted phoneme tokens within their annotated temporal boundaries. These metrics serve as objective proxies for intelligibility [29] and phonetic reconstruction. They do not replace listening tests but provide a reproducible evaluation across conditions. Metrics are computed on the left ear [25]. For all metrics except for the WER, higher is better. Statistical significance is assessed via the Wilcoxon signed-rank test [30] performed between each method against the baseline, at a level of $\alpha = 5\%$. Significant gains are highlighted in bold fonts in the tables.

[1] The source code for this work is publicly available at `https://github.com/Nasseredd/fw-se-loss`

Table 2: *Performance for phoneme categories (consonants and vowels) for the WN and SSN test sets averaged across input* SIR *levels.*

	Loss	WN							SSN						
		SIR	SAR	SDR	FW-SIR	FW-SAR	FW-SDR	PA	SIR	SAR	SDR	FW-SIR	FW-SAR	FW-SDR	PA
Consonants	$\mathcal{L}_{\mathrm{T}}$	12.8	**2.0**	**1.4**	6.5	**3.9**	6.3	34.0	14.7	**0.8**	**0.5**	9.9	**2.4**	**6.4**	43.6
	$\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$	**16.1**	1.4	1.1	**9.7**	3.0	**6.7**	**36.1**	14.6	0.4	0.0	9.8	2.1	6.1	**45.5**
Vowels	$\mathcal{L}_{\mathrm{T}}$	14.4	**3.2**	**2.7**	11.4	**4.6**	**8.4**	43.7	13.8	**1.9**	**1.3**	12.5	3.0	7.6	46.4
	$\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$	**17.8**	2.7	2.5	**14.8**	3.9	7.7	**45.5**	**15.1**	1.2	0.8	**13.8**	**3.3**	**8.4**	**47.9**

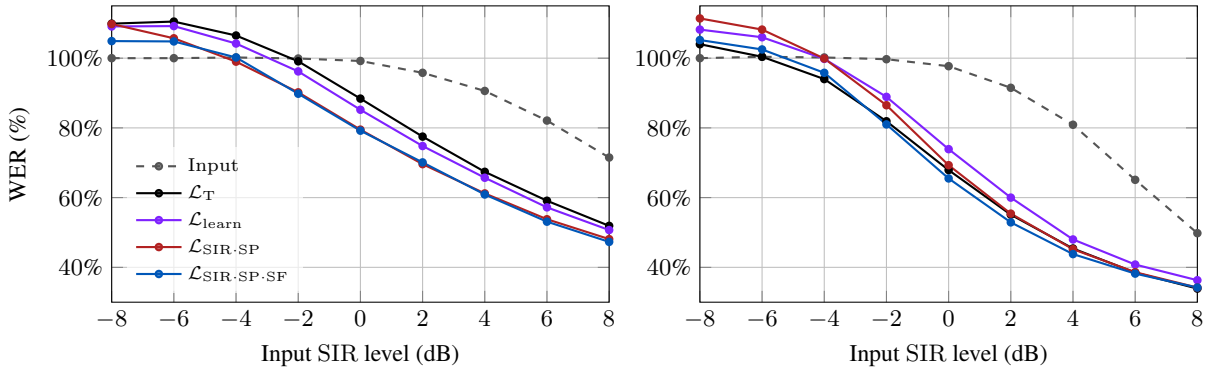


Figure 1: WER *(%) as a function of the input* SIR *level for WN (left) and SSN (right) test sets.*

4. Results

4.1. Results at the utterance level

Table 1 displays utterance-level performance in terms of classical, frequency-weighted, and intelligibility metrics. First, we observe that under WN, all losses yield an improved SIR and FW-SIR, with the $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ achieving the strongest overall improvement. Introducing the SIR-based weighting or learning frequency weights does not lead to a substantial degradation in SAR, FW-SAR, SDR, FW-SDR, or STOI, as results remain overall comparable across losses, with only slight variations relative to the baseline.

Under SSN, both $\mathcal{L}_{\mathrm{learn}}$ and $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ slightly improve interference reduction compared to the baseline. In contrast, $\mathcal{L}_{\mathrm{logSIR}}$ and $\mathcal{L}_{\mathrm{SIR\cdot SP}}$ do not consistently reduce interference. On the other hand, SAR and SDR slightly decrease for the weighted losses, with a more pronounced reduction for $\mathcal{L}_{\mathrm{logSIR}}$ and a smaller one for $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$. Frequency-weighted metrics remain overall comparable to the baseline, with a more noticeable drop for $\mathcal{L}_{\mathrm{SIR\cdot SP}}$, whereas $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ achieves the highest FW-SAR and FW-SDR scores. Finally, STOI decreases for all weighted losses under SSN, particularly for $\mathcal{L}_{\mathrm{logSIR}}$ and $\mathcal{L}_{\mathrm{SIR\cdot SP}}$, while the drop remains more limited for $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$.

Overall, these results indicate that weighting strategies combining interference awareness with additional spectral or temporal structure—either through learned frequency profiles or the spectral-flux–augmented scheme—yield more consistent interference suppression across noise types, whereas relying solely on local SIR-based time–frequency modulation, such as $\mathcal{L}_{\mathrm{logSIR}}$ and $\mathcal{L}_{\mathrm{SIR\cdot SP}}$, exhibits less stable performance, particularly under spectrally shaped noise.

We now examine speech recognition results in terms of WER, as shown in Fig. 1, focusing on the three proposed weighting schemes. For WN, despite being relatively high, the gap between input and output WER is consistent with observations reported when comparing speech recognition-based evaluation with listening tests [31]. All weighted losses yield a lower WER than the baseline across most SIR levels. While $\mathcal{L}_{\mathrm{learn}}$ yields consistent improvements, both $\mathcal{L}_{\mathrm{SIR\cdot SP}}$ and $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ significantly decrease the WER, particularly at mid-to-high SIR levels. In the SSN condition, trends are less consistent. The loss $\mathcal{L}_{\mathrm{learn}}$ performs worse than the baseline across all SIR levels. Below 0 dB, the baseline achieves the lowest WER, whereas above 0 dB both SIR-based losses perform similarly to the baseline, with a slight advantage for $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ at mid-to-high SIR levels. Overall, these results confirm previous observations: interference-aware weighting improves recognition performance, particularly under WN conditions. However, incorporating additional spectral or temporal structure, as in $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$, provides the most robust and consistent behavior.

4.2. Performance on consonants and vowels

We now report the various performance metrics computed for phoneme categories in Table 2. For clarity, we focus on comparing the proposed $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ loss against the $\mathcal{L}_{\mathrm{T}}$ baseline.

For consonants under WN, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ improves interference suppression, as reflected by higher SIR and FW-SIR compared to the baseline, while maintaining comparable FW-SDR and improving PA. Although SAR and SDR slightly decrease relative to the baseline, the drop remains limited, indicating that improved interference suppression does not induce substantial additional distortion. Under SSN, interference reduction in terms of SIR and FW-SIR remains comparable to the baseline. Slight reductions are observed in artifact-related and overall distortion metrics, both classical and frequency-weighted. Nevertheless, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ results in an improvement in PA.

For vowels under WN, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ increases both SIR and FW-SIR, while showing slight reductions in artifact- and distortion-related metrics. Despite these minor decreases, PA improves compared to the baseline, indicating that the enhanced interference suppression benefits vowel recognition. Under SSN, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ achieves higher SIR and frequency-weighted

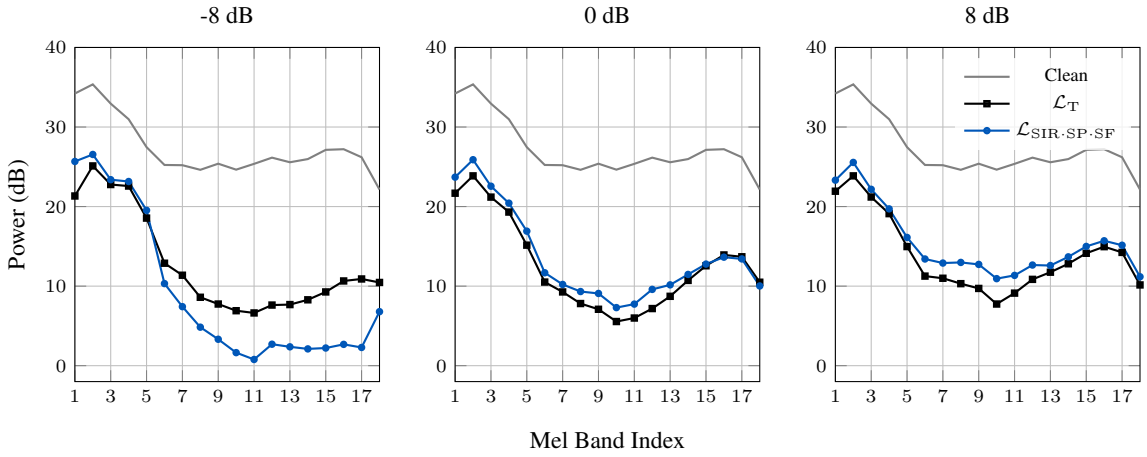


Figure 2: *Comparison of estimated plosive spectra across* -8 *dB,* 0 *dB, and* 8 *dB input* SIR *under* WN. *The plots illustrate the spectral reconstruction performance of* $\mathcal{L}_{\mathrm{T}}$ *and* $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ *relative to the clean reference spectrum.*

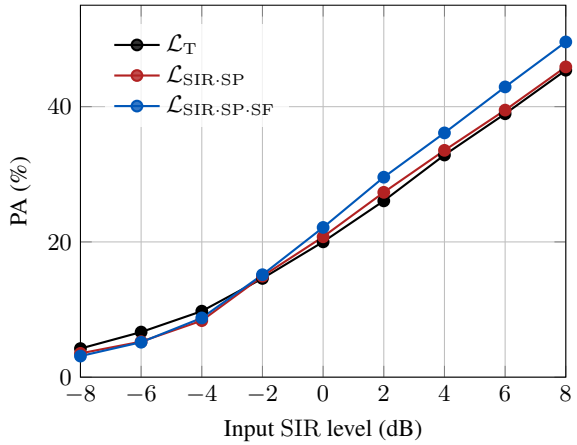


Figure 3: *Plosive* PA *under* WN *across input* SIR *levels.*

metrics than the baseline, together with the highest vowel PA. Despite moderate variations in SAR and SDR, the overall performance remains stable.

These phoneme-level results are consistent with the utterance-level findings: $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ tends to provide more consistent improvements across phoneme classes, particularly under WN conditions.

4.3. Plosive accuracy across SIR levels

To further analyze phoneme-level performance observed in Table 2, we examine plosive recognition across SIR levels under WN (Fig. 3). Plosives are characterized by brief transients and rapid spectral changes, making them particularly sensitive to masking and well suited to assess the proposed spectral-flux-based weighting.

Across increasing SIR levels, all losses exhibit a monotonic improvement in plosive PA. At very low SIR, performance remains similar, with the baseline slightly higher. However, from mid SIR levels onward, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ significantly outperforms the baseline from 0 dB and $\mathcal{L}_{\mathrm{SIR\cdot SP}}$ from 2 dB onward, with the performance gap widening at higher SIR levels.

This trend confirms that incorporating SIR-aware and spectral-flux-based weighting enhances recognition of transient phonemes such as plosives, particularly in moderate-to-high SIR conditions where interference suppression becomes more effective.

4.4. Spectral analysis of plosives

To better understand the spectral behavior underlying the plosive results, Fig. 2 compares the estimated spectra to the clean reference across -8, 0, and 8 dB under WN in the Mel scale.

At -8 dB, both methods remain far from the clean spectrum. Up to band 6, the two estimates are similar. Beyond band 6, $\mathcal{L}_{\mathrm{T}}$ remains closer to the clean reference, while $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ shows larger deviation in the mid-to-high bands. At 0 dB, $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ becomes slightly closer to the clean spectrum than the baseline, particularly across bands 8–13, where it better tracks the clean spectral shape. At 8 dB, the same tendency is observed with a larger margin, especially over bands 6–12, where $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ more closely follows the clean spectrum.

Overall, the benefit of $\mathcal{L}_{\mathrm{SIR\cdot SP\cdot SF}}$ in reconstructing the mid-frequency structure of plosives is mainly observed at moderate and high input SIR levels. At very low SIR, the baseline remains closer to the clean spectrum in several mid-to-high frequency bands. These less adverse listening conditions are particularly relevant for hearing-aid applications.

5. Conclusion

We proposed a TF weighting framework for SDR-based speech enhancement that explicitly models speech presence, local SIR, and transient dynamics. The proposed formulations reshape the contribution of TF bins according to speech–noise competition, while preserving the stability of SDR optimization. Experimental results show that incorporating SIR-aware and spectral-flux-based modulation improves interference suppression and phoneme-level reconstruction. Future work will explore additional perceptually-based phenomena, such as frequency-dependent speech reception thresholds, band-specific masking, or intelligibility-weighted filters. Such directions may better align training with phoneme-level perception and hearing-assistive objectives.

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7. Generative AI Use Disclosure

We acknowledge the ISCA policy regarding the use of generative AI tools. The authors declare that generative AI tools have been used solely to correct grammar. No such tools were used to write significant parts of this manuscript.

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